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Part – A : Conceptual foundation

(1) What is Conditional Probability?

→ Conditional probability is the probability of an event occurring given that another event has already occurred. It helps measure the likelihood of one event under a specific condition.

$$P(A/B) = \frac{P(A \cap B)}{P(B)}$$

(2) Explain Bayes theorem and its importance in classification problems.

→ Bayes Theorem calculates the probability of a class bassed on prior knowledge and observed data. It is widely used in the Naive Bayes classifier to predict whether a message belongs to the spam or legitimate class.

$$P(A/B) = \frac{P(B/A) \times P(A)}{P(B)}$$

(3) What assumptions does the Naive Bayes Classifier?

→ Naive Bayes assumes that all input features are conditionally independent given the class label. This means each features contributes independently to the prediction, even if they are correlated in reality.

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(4) (a) K-Nearest Neighbors

→ KNN is a distance-based supervised learning algorithm. It classifies a new data point by finding the KNN and assigning the class with the majority vote. It commonly uses Euclidean distance to measure similarity.

(b) Support Vector Machine

→ SVM is a supervised learning algorithm that finds the optimal hyperplane to separate different classes with the maximum margin. It uses SVM to define the decision boundary and can handle non linear data using kernel functions such as the RBF.

(5) Compare distance-based, probabilistic and margin based classifiers.

Type	Example	Principle
Distance Based	KNN	Classifies data based on the ~~K~~ KNN samples.
Probabilistic	Naive Bayes	Uses Bayes theorem to calculate class probabilities
Margin Based	SVM	Separates classes using the maximum-margin hyperplane